Mijeong Kim

Ph.D. Student in Computer Vision Lab, SNU

Personal info

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Research Interest

Computer Vision, 3D Vision, NeRFs, Human/Scene Reconstruction, Deep Learning

- Recently, I am interested in NeRF based 3D GAN and Few-Shot 3D reconstruction, but opened to other subjects.

Education

Seoul National University, Korea 2019 - in progress
PhD student, Computer Vision Lab, Dept. of EE

- Advisor: Prof. Bohyung Han

Seoul National University, Korea 2014 - 2019
BS, Naval Architecture and Ocean Engineering

- Valedictorian and Summa Cum Laude

Gyeongnam Science High School, Korea 2012 - 2014

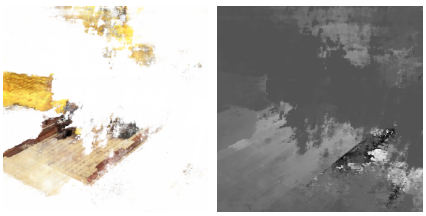
- Entering with the 1st place, and Early Graduation (3Y→2Y), Major in Physics

My strengths in 3D Research

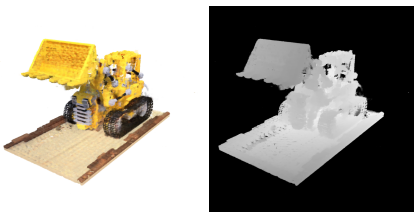
Since I studied mechanics (including dynamics, statics, fluid mechanics) and graphics in undergraduate/graduate courses and physics olympiad in high school, I am familiar with 3D coordinates, camera rotations, refraction, reflections, etc. My experience would be competitive to conduct research on 3D Vision, including reconstruction, rendering, etc.

Publication

1. Mijeong Kim, Seonguk Seo, Bohyung Han, "InfoNeRF: Ray Entropy Minimization for Few-Shot Neural Volume Rendering", **CVPR2022** [Paper] [Project]



(a) NeRF in 4-view setting



(b) InfoNeRF(ours) in 4-view setting

Other Research Experiences & Projects

1. Mijeong Kim*, Geeho Kim*, Minji Kim*, "Overcoming Forgetting in Federated Learning via Importance From Agent" [Paper] [Code]
   - Conducted in 'Machine Intelligence and Data Science Lab. (430.658 of SNU, Prof. Taesup Moon)', 2021
2. Mijeong Kim*, Donghwan Jang, Bohyung Han, "Dynamic Spatially-Adaptive Modulation for Image Dehazing", 2021 [Paper]
3. Mijeong Kim*, Sanghyeok Chu*, Bohyung Han, "Beyond Homography: Nonparametric Image Alignment via Graph Convolutional Networks", 2020 [Paper]
4. Mijeong Kim, OpenGL Projects (Skeletal animation, Lighting etc) [Video] [Paper]
   - Conducted in '3D Vision Lab. (430.638 of SNU, Prof. Youngmin Kim)', 2020

(* denotes equal contributions)

Awards & Honors

Kwanjeong Ph.D. Fellowship 2019 - 2021
Full Funding. One of the prestigious Ph.D. scholarships in Korea

Seoul National University Tomorrow's Engineers Membership 2016-2018
Student Honor Society, selecting students with outstanding academic abilities in SNU.

Science and Engineering Scholarship by President 2014 - 2018
Full Funding. The most prestigious scholarship in the South Korea

Talent Medal of Korea by President 2014
Physics area. Recognition award by president of Korea.

Teaching Assistant

- Samsung Electronics, Korea — Jul 2019 & Jul 2020
- Hyundai Motors, Korea — Mar 2020 - Jun 2020
- Digital Computer Concept and Practice — Spring 2020
  Introduction to Python, Dept. of EE in SNU

Grants

Deep Learning-Based Pedestrian Removal with Multi-View Data

- Fund by Naver — Sep 2019 - Aug 2020

Invited Presentations

- Samsung Electronics, Korea — Jun 2021